

# Multiple integrals – 4

Line and surface integrals: Green, Stokes, Gauss

Homework

13 сентября 2026 года

**Theoretical minimum.** The three classical formulas are one statement,  $\int_{\partial M} \omega = \int_M d\omega$ , in dimensions one, two and three: Newton–Leibniz, Green and Stokes, Gauss–Ostrogradsky. Each of them converts a boundary integral into an integral of a density, so an area is computed by a contour integral and a flux by an integral of the divergence. A field is a gradient exactly when it is closed and the domain is simply connected; on a domain with a hole the obstruction is measured by the integral over a loop, that is by the winding number. The Newtonian field  $x/\|x\|^3$  is divergence free away from the origin and has flux  $4\pi$  through any surface enclosing it — every statement about sources and singularities is a variant of this example. Finally, Green’s identities turn the Laplacian into boundary data and give the mean value property of harmonic functions.

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1. Prove that a closed  $C^1$  field on a simply connected domain has a potential, and show that  $\frac{(-y, x)}{x^2+y^2}$  on  $\mathbb{R}^2 \setminus \{0\}$  is closed but has none; compute its integral over a loop and interpret the answer as the winding number.
2. Prove that the winding number of a closed curve about a point is an integer, that it is locally constant in the point, and that it vanishes on the unbounded component of the complement.
3. Compute  $\oint_C \frac{x dy - y dx}{x^2 + y^2}$  for a smooth closed curve  $C$  not passing through the origin, in terms of the winding number, and deduce the argument principle for a rational function (compare with Complex analysis – 3).

4. Prove the first and the second Green identities, and deduce from them the mean value property of a harmonic function and the uniqueness of the solution of the Dirichlet problem.
5. Compute the area of the surface of revolution generated by a smooth curve, prove Archimedes' theorem that the sphere and the circumscribed cylinder have equal lateral areas between two parallel planes, and deduce the uniform distribution of the projection of the sphere onto an axis.
6. Compute the flux of the field  $F = (x^3, y^3, z^3)$  through the unit sphere, and  $\iint_S \langle x, d\vec{S} \rangle$  for an arbitrary closed surface, interpreting the answer as the volume.
7. Let  $F$  be a smooth field on  $\mathbb{R}^3$  with  $\operatorname{div} F = 0$ . Prove that  $F$  is a curl of another field, construct that field explicitly, and explain what changes on  $\mathbb{R}^3 \setminus \{0\}$ .
8. Prove the isoperimetric inequality  $4\pi A \leq L^2$  for a smooth simple closed plane curve by Green's formula and Wirtinger's inequality, and compare with the proofs in Convex – 4 and in Fourier analysis – 2.
9. Let  $u$  be harmonic in a neighbourhood of the closed unit ball in  $\mathbb{R}^3$ . Prove that  $\iint_{S^2} u \, d\sigma = 4\pi u(0)$  by the divergence theorem, and deduce the maximum principle.
10. Prove the change of variables formula in the plane from Green's theorem: for a diffeomorphism  $\varphi$  of a region show that  $\iint_{\varphi(D)} dx \, dy = \iint_D |\det D\varphi|$ , and explain why the sign of the Jacobian corresponds to the orientation of the boundary.